

Identities, Multiple Angles, and Exact Solutions

Answer Key

Precalculus

Quick Answers

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|------------------------------------|--|
| 1. $\sin(x)$ | 7. $0, \frac{2\pi}{3}, \frac{4\pi}{3}$ |
| 2. $\cos(x)$ | 8. $\tan(x)$ |
| 3. $1 - 2\sin^2(x)$ | 9. $2\sec^2(x)$ |
| 4. $\frac{\pi}{6}, \frac{5\pi}{6}$ | 10. $0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}$ |
| 5. $\frac{24}{25}$ | 11. $-\frac{119}{169}$ |
| 6. $2\sin(x)$ | 12. $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ |
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Common wrong answers

5. If they got 1.2: doubled the sine instead of using the double-angle formula
11. If they got -0.77: doubled the cosine instead of using the double-angle formula
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1. Simplify $\frac{1 - \cos^2 x}{\sin x}$.

Step 1. The numerator is one of the two rearrangements of the Pythagorean identity, so replace it rather than expanding anything.

Step 2. $1 - \cos^2 x = \sin^2 x$, so the expression is $\frac{\sin^2 x}{\sin x}$.

Step 3. Cancel one factor of $\sin x$: $\boxed{\sin x}$ (valid where $\sin x \neq 0$).

2. Simplify $\sec x - \sec x \sin^2 x$.

Step 1. Both terms share $\sec x$; factoring first exposes a Pythagorean form instead of leaving two unrelated terms.

Step 2. $\sec x(1 - \sin^2 x) = \sec x \cos^2 x$.

Step 3. $\sec x = \frac{1}{\cos x}$, so $\frac{\cos^2 x}{\cos x} = \boxed{\cos x}$.

3. Rewrite $\cos 2x$ **in terms of** $\sin x$ **alone.**

Step 1. There are three forms of the cosine double angle. Choosing is the whole point: pick the one whose surviving function is the one you were asked for.

Step 2. From $\cos 2x = \cos^2 x - \sin^2 x$, substitute $\cos^2 x = 1 - \sin^2 x$: $(1 - \sin^2 x) - \sin^2 x$.

Step 3. $\cos 2x = \boxed{1 - 2\sin^2 x}$

4. Solve $2 \sin x - 1 = 0$ **on** $[0, 2\pi)$.

Step 1. One trigonometric function, first power — isolate it and read the unit circle; no identity is needed here.

Step 2. $\sin x = \frac{1}{2}$.

Step 3. Sine is positive in quadrants I and II. The reference angle is $\frac{\pi}{6}$, so the quadrant II partner is $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$.

Step 4. $x = \frac{\pi}{6}, \frac{5\pi}{6}$

5. $\sin \theta = \frac{3}{5}$ in quadrant I; find $\sin 2\theta$.

Step 1. A double angle with both $\sin \theta$ and $\cos \theta$ known calls for $\sin 2\theta = 2 \sin \theta \cos \theta$ — no quadrant work is left, since quadrant I makes both values positive.

Step 2. $2 \left(\frac{3}{5}\right) \left(\frac{4}{5}\right) = \frac{24}{25}$.

Step 3. $\sin 2\theta = \frac{24}{25}$

If they got 1.2: doubled the sine itself. $\sin 2\theta$ is never $2 \sin \theta$ — that value even exceeds the range of sine.

6. Simplify $\tan x \cos x + \sin x$.

Step 1. A lone tangent among sines and cosines is the signal to rewrite it as a quotient; nothing else in the expression needs touching.

Step 2. $\tan x \cos x = \frac{\sin x}{\cos x} \cos x = \sin x$.

Step 3. The expression is $\sin x + \sin x = 2 \sin x$.

7. Solve $2 \cos^2 x - \cos x - 1 = 0$ **on** $[0, 2\pi)$.

Step 1. One function appears, but squared — so this is a quadratic in $\cos x$. Factor rather than isolating.

Step 2. $(2 \cos x + 1)(\cos x - 1) = 0$, giving $\cos x = -\frac{1}{2}$ or $\cos x = 1$.

Step 3. $\cos x = -\frac{1}{2}$ has reference angle $\frac{\pi}{3}$ and is negative in quadrants II and III, giving $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$. $\cos x = 1$ gives $x = 0$.

Step 4. $x = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$

8. Simplify $\frac{\sin 2x}{1 + \cos 2x}$.

Step 1. Both double angles must go, and the denominator decides which cosine form to use: $\cos 2x = 2 \cos^2 x - 1$ is the one that makes $1 + \cos 2x$ collapse.

Step 2. Numerator: $\sin 2x = 2 \sin x \cos x$. Denominator: $1 + (2 \cos^2 x - 1) = 2 \cos^2 x$.

Step 3. $\frac{2 \sin x \cos x}{2 \cos^2 x} = \frac{\sin x}{\cos x} = \tan x$

9. Combine $\frac{1}{1 - \sin x} + \frac{1}{1 + \sin x}$.

Step 1. The two denominators are conjugates, so their product is a difference of squares — which is exactly the shape the Pythagorean identity simplifies.

Step 2. Common denominator $(1 - \sin x)(1 + \sin x) = 1 - \sin^2 x = \cos^2 x$; the numerators add to $(1 + \sin x) + (1 - \sin x) = 2$.

Step 3. $\frac{2}{\cos^2 x} = \boxed{2 \sec^2 x}$

10. Solve $\sin 2x = \sin x$ **on** $[0, 2\pi)$.

Step 1. The two sides carry different angles, so expand the double angle first. Then move everything to one side — dividing by $\sin x$ would destroy the solutions where $\sin x = 0$.

Step 2. $2 \sin x \cos x - \sin x = 0$, so $\sin x(2 \cos x - 1) = 0$.

Step 3. $\sin x = 0$ gives $x = 0$ and $x = \pi$. $\cos x = \frac{1}{2}$ gives $x = \frac{\pi}{3}$ and $x = \frac{5\pi}{3}$.

Step 4. $\boxed{x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}}$

11. $\cos \theta = -\frac{5}{13}$ in quadrant II; find $\cos 2\theta$.

Step 1. Only the cosine is known, so choose the double-angle form built from cosine alone: $\cos 2\theta = 2 \cos^2 \theta - 1$. The quadrant does not matter here, because the cosine is squared.

Step 2. $2 \left(-\frac{5}{13}\right)^2 - 1 = 2 \cdot \frac{25}{169} - 1 = \frac{50}{169} - \frac{169}{169}$.

Step 3. $\cos 2\theta = \boxed{\frac{-119}{169}}$

If they got -0.77 : doubled the cosine instead of squaring it first.

12. Challenge: solve $2 \cos^2 x + \sin x = 1$ **on** $[0, 2\pi)$.

Step 1. Two different functions appear and one is squared, so the first move is an identity that leaves a single function — replace $\cos^2 x$ with $1 - \sin^2 x$ so everything is in sine.

Step 2. $2(1 - \sin^2 x) + \sin x = 1$, so $2 - 2 \sin^2 x + \sin x - 1 = 0$, i.e. $2 \sin^2 x - \sin x - 1 = 0$.

Step 3. Factor: $(2 \sin x + 1)(\sin x - 1) = 0$, so $\sin x = -\frac{1}{2}$ or $\sin x = 1$.

Step 4. $\sin x = 1$ gives $x = \frac{\pi}{2}$. $\sin x = -\frac{1}{2}$ is negative in quadrants III and IV, with reference angle $\frac{\pi}{6}$, giving $\frac{7\pi}{6}$ and $\frac{11\pi}{6}$.

Step 5. $\boxed{x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}}$